contains carbon, hydrogen and oxygen, but it differs from a carbohydrate in that it contains more carbon and hydrogen and less oxygen. When oxidized, it gives nine kilocalories. A molecule of fat consists of three molecules of fatty acids and one molecule of glycerol. It is also known as *triglyceride*.

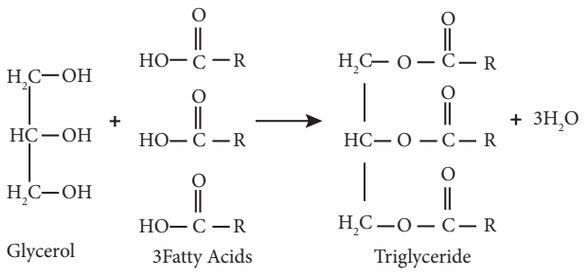


Fig 10.11: Formation of Triglyceride

10.9 Differences between fats and oils

Fats and oils have many types of triglycerides. Nature of fat or oil depends on the type of fatty acids attached to the glycerol molecule.

Table 10.6: Difference between Fats and Oils

S.no	Fats	Oils
1	Fat is solid at room temperature	Oil is liquid at room temperature
2	Fat is obtained from animals (i.e) butter, ghee, egg yolk ,meat	Oils are obtained from plant sources (i.e) mustard oil, groundnut oil, almond oil
3	Fats have only saturated fatty acids	Oils have saturated and unsaturated fatty acids

DO YOU KNOW...?

- Fats are a concentrated source of energy
- Good fats can promote health and bad fats can increase risk of chronic diseases



Fig 10.12: Fats and oils

10.10 Classification of fats

Fats are classified into 4 categories as follows:

I. On the basis of chemical composition
II. On the basis of fatty acids
III. On the basis of requirement
IV. On the basis of sources

10.10.1 On the basis of chemical composition

Fats can be classified into 3 main groups as follows:

1) Simple lipids

These are esters of fatty acids and glycerol. They are also called as neutral fats or triglycerides. These neutral fats make up 98 -99% of food and body fats. (e.g) fats and oils

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